

flea

NEWS 95 *April 2025*



FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

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Siphonaptera Literature (2024)

(includes duplicates from FN94)

New Flea Species

(2024)

[*Strepsylla mohinora* sp. nov. Acosta et al. 2024]

[(*Ctenidiosomus eckerlini* Hastriter 2024, *Ctenidiosomus churchi* Hastriter 2024, *Ctenidiosomus pattersoni* Hastriter 2024, and *Ctenidiosomus solusensus* Hastriter 2024]

[*Nosopsyllus (Nosopsyllus) spiniformis* n. sp. Keskin 2024]

(abstracts in previous FN)

Oliveira, T., da Costa, J. P., Araújo, A. V., Dias, D. E. M., & Dantas, M. A. T. (2024). Mangy armadillos: New records of *Karethraichnus minimum* in extinct quaternary pampatheres of the Brazilian intertropical region. *Journal of South American Earth Sciences*, 144, 105034.

This paper presents pathological lesions in the osteoderms of three extinct pampatheres: two specimens of the genus *Holmesina* sp. and one specimen of *Pampatherium humboldtii*, which lived in the Brazilian Intertropical Region. The traces analyzed here were attributed to the ichnospecies *Karethraichnus minimum* and were associated with the behavior of ectoparasites, such as fleas, likely belonging to the genus *Tunga*. Some osteoderms also present ulcerative lesions, possibly generated by the action of microorganisms such as bacteria and fungi. These new records expand our knowledge about the paleopathologies caused by fleas in extinct pampatheriids, demonstrating that such occurrences were common.

John-Paul Zonneveld, Oscar Wilson, Patricia Holroyd. 2024. Composite ichnological-pathological evidence for arthropod parasitism on osteoderms of *Boreostemma acostae* (Glyptodontidae, Cingulata) from La Venta, Colombia, *Journal of South American Earth Sciences*, Volume 146, 2024, 105085, ISSN 0895-9811, <https://doi.org/10.1016/j.jsames.2024.105085>.

[ichnospecies *Karethraichnus kulindros*, *K. minimum*, *K. zaratan*]

[These are possibly *Tunga*. If *Tunga perforans*, would they be synonyms? Editor]

Zhou, S., Liu, W., Zeng, Z., Wang, J., Han, T., Xu, G., ... & Xiao, F. (2024). Taxonomy and phylogeny of rodents parasitic fleas in southeastern China with description of a new subspecies of *Ctenophthalmus breviprojiciens*. *Scientific Reports*, 14(1), 28316.

Accurate differentiation and identification of flea species are essential for both basic and applied research on fleas, as well as for the diagnosis of flea-borne diseases. However, distinguishing between flea species can be challenging, especially among those with minimal morphological differences. Therefore, some scholars have suggested the necessity of comprehensive revisions to the classification of fleas, incorporating morphological, molecular, and phylogenetic data. In this study, we focused on classifying the rodents' parasitic fleas in southeastern China and provided molecular and phylogenetic data. We also described a new subspecies *Ctenophthalmus breviprojiciens fujiansis* n. ssp. A total of 392 fleas were collected from 8 species of rodents in 10 counties. Morphologically, they belonged to 10 species, 9 genera and 5 families. Barcode identification based on COI gene and phylogenetic analysis based on five genetic markers (18S rDNA, 28S rDNA, EF-1a, COI, COII) revealed that the molecular and morphological identification of *Xenopsylla cheopis*, *Aviostivalius klossi bispiniformis*, *Leptopsylla segnis*, *Monopsyllus anisus* and *Ctenocephalides felis felis* were consistent. The taxonomic status of *Neopsylla specialis minpiensis* and *Peromyscopsylla himalaica sinica* as subspecies is questionable due to significant intraspecific genetic distance, and further morphological and molecular data are required to determine if they should be elevated to species level. The molecular identification of *C. breviprojiciens* n. ssp., *N. dispar fukienensis*, and *Nosopsyllus nicanus* could not be completed at this time due to a lack of sequences for related species in existing GenBank databases. Additionally, phylogenetic relationships of 31 species from 9 genera and 5 families of Siphonaptera were inferred based on five molecular markers (18S rDNA, 28S rDNA, EF-1a, COI and COII) using Maximum Likelihood analyses. The analyses revealed that various taxa of Siphonaptera are monophyletic at the subspecies, species, and genus levels. However, at the family level, Leptopsyllidae, Ceratophyllidae, Pulicidae, and Pygiopsyllidae are all monophyletic, while Ctenophthalmidae is paraphyletic. we support the view of some authors that revising the catchall group Ctenophthalmidae and elevating each of its constituent subfamilies to family status.

[*Ctenophthalmus breviprojiciens fujiansis* n. ssp.]

BAI Xue-l,* GAO Xue-ping, LU Liang, WEI Hao.

白学礼, 高学平, 鲁亮, & 魏浩. (2024). 中国狭臀蚤属一新种记述 (蚤目, 栉眼蚤科). *寄生虫与医学昆虫学报*, 31(01), 42-45.

A NEW SPECIES OF *STENISCHIA* FROM NINGXIA, CHINA (SIPHONAPTERA, CTENOPHTHALMIDAE)

Acta Parasitol. Med. Entomol. Sin.

doi:10.3969/j.issn.1005-0507.2024.01.007 中图分类号: R384.3

Bai Xueli, Gao Xueping, Lu Liang, & Wei Hao. (2024). Description of a new species of the genus

STENISCHIA from China (Siphonaptera, CTENOPHTHALMIDAE). Acta Parasitologica Sinica, 31(01), 42-45.

记述栉眼蚤科 Ctenophthalmidae 狭臀蚤属 *Stenischia* Jordan, 1932 的 1 新种, 宁夏狭臀蚤 *Stenischia ningxiaensis* Bai, Lu et Gao, sp. nov., 标本采自宁夏南部地区。模式标本保存于军事科学院

The present paper deal with a new species *Stenischia* Jordan, 1932 from Ningxia. Types specimens were collected from Ningxia. Alltypes are deposited in the Medical Entomology Collection of Academy of Military Medical Science, Beijing, China and Ningxia Hui Autonomous Region Center for Disease Control and Prevention.

[*Stenischia ningxiaensis* Bai, Lu et Gao 2024, sp. nov.], (Figs. 1-13)

Taxonomy

Curtiss, R. T., & Needham, K. M. (2024). Primary insect type specimens in the University of British Columbia–Beaty Biodiversity Museum–Spencer Entomological Collection. Journal of the Entomological Society of British Columbia, 121, e2613-e2613.

Hastriter, M. W., & Bossard, R. L. (2025). Robert E. Lewis World Species Flea (Siphonaptera) List (version 2.0). In O. Bánki, Y. Roskov, M. Döring, G. Ower, D. R. Hernández Robles, C. A. Plata Corredor, T. Stjernegaard Jeppesen, A. Örn, T. Pape, D. Hobern, S. Garnett, H. Little, R. E. DeWalt, K. Ma, J. Miller, T. Orrell, R. Aalbu, J. Abbott, R. Adlard, et al., Catalogue of Life (Version 2025-03-14). Catalogue of Life, Amsterdam, Netherlands. <https://doi.org/10.48580/dfvll-48fy>

Di Lorenzo, N., Mei, M., Ascenzi, A., Cerretti, P., Chimenti, C., & Badano, D. (2024). The flea collection of the Museum of Zoology, Sapienza University of Rome (Insecta: Siphonaptera). Fragmenta Entomologica, 56(2).

Jiang, L., Chen, J., Li, Z., Nan, Y., & Qiao, G. (2024). A report on newly described taxa of Hemiptera and other 28 orders of Insecta in China (2022–2023). Biodiversity Science, 32(11), 24411.

Marcela Lareschi. 2024. Chapter 63 in Concepts in Animal Parasitology on fleas, order Siphonaptera, by S. L. Gardner and S. A. Gardner, editors. Zea Books, Lincoln, Nebraska, United States. doi: 10.32873/unl.dc.ciap063

Lewis, R. E., & Eckerlin, R. P. (2024). The Siphonaptera of North America north of Mexico, including Greenland. Annals of Carnegie Museum, 89(1), 1-562.

Four major books were published on the flea fauna of North America during the 20th Century. The first of these was on the fleas of the eastern United States by Irving Fox (1940a). The second, dealing with the fleas west of the 100th meridian in the United States was by C. Andresen Hubbard (1947). The third and fourth concern the Canadian flea fauna and both were written by George P. Holland (1949b and 1985). The latter of these included Greenland and Alaska as well. Many new taxa have been described and many taxonomic changes have occurred since the publication of the older works. The current treatment addresses the fleas of all of North America north of Mexico. A chapter on flea development is followed by a description of the anatomy of adult fleas. The major focus of the volume is the identification of taxa. There are keys to the identification of the ten families, 80 genera,

and 316 species of North American fleas. For each species, there are illustrations of the taxonomically important features of both the male and female adults. Host and biogeographic information are given for each species. Extensive lists of the published reports of the species are designed to aid future workers. Appendices address biographies of North American flea workers, lists of new synonymies, and a host list by current scientific name. A taxonomic index is provided for the fleas along with a separate index for the hosts.

[Price: \$10.00 for an electronic copy (pdf),

<https://bioone.org/journals/annals-of-carnegie-museum/volume-89/issue-1/007.089.0102/The-Siphonaptera-of-North-America-North-of-Mexico-Including-Greenland/10.2992/007.089.0102.short>

Reportedly available in print, too. Contact: Editor, bossardtech@gmail.com]

Linardi, P. M. (2024). Capítulo 35: Siphonaptera Latreille, 1825. Insetos do Brasil: Diversidade e Taxonomia. 2ª ed.

Vanin, S. A., Nihei, S. S., & Souza-Dias, P. G. (2024). Capítulo 3: Filogenia e classificação. Insetos do Brasil: Diversidade e Taxonomia. 2ª ed.

Constantino, R. (2024). Capítulo 5: A importância dos insetos. Insetos do Brasil: Diversidade e Taxonomia. 2ª ed.

Machado, R. J., & Collucci, E. (2024). Capítulo 34: *Mecoptera* Packard, 1886. Insetos do Brasil: Diversidade e Taxonomia. 2ª ed.

Constantino, R., & Aguiar, A. J. (2024). Capítulo 37: Glossário. Insetos do Brasil: Diversidade e Taxonomia. 2ª ed.

Rafael, J. A., Melo, G. A., de Carvalho, C. J., Casari, S. A., & Constantino, R. (2024). Apresentação: A diversidade de insetos no Brasil. Insetos do Brasil: Diversidade e Taxonomia. 2ª ed.

Mathison, B. A. (2024). Arthropods of Medical Importance. ClinMicroNow, 1-24.

Medvedev, S. G. (2024). Morphological basis of the three classifications of the order of fleas (Insecta, Siphonaptera). Parazitologiya, 58(2), 147-168.

The morphological basis and the reasons of differences between three classifications of the order of fleas (Siphonaptera), proposed in the second half of the 20th century, are analyzed.

Keywords: fleas, Siphonaptera, classifications, morphology, character states

Porsani, L. N. (2024). Estudo de Zoologia por comparação: criação de guias ilustrados. [Study of Zoology by comparison: creation of illustrated guides.] [by using illustrated dichotomous keys.]

Sawada, Y., Minei, R., Tabata, H., Ikemura, T., Wada, K., Wada, Y., ... & Iwasaki, Y. (2024). Unsupervised AI reveals insect species-specific genome signatures. PeerJ, 12, e17025.

Sazmand, A., Amor, N., Hastriter, M., Zhang, Q., Wu, L., Yang, X., ... & Zhang, Q. (2024). OPEN

ACCESS, Wildlife Parasitology: Emerging Diseases and Neglected Parasites, 57. (n.p.): Frontiers Media SA.

Semuel, M. Y. (Ed.). (2024). EKTOPARASIT GENUS *CTENOCEPHALIDES* (Diversitas Genetic, Barcoding Molekuler, Bakteri Pathogen dan Bioinsektisida). Lakeisha.

Zhao, C., Huang, M., Yang, D., & Liu, X. (2024). Comparative morphology of the wing base structure illuminates higher-level phylogeny of Holometabola. *Insects*, 15(3), 199.

Conservation

Colombo, C., Pontani, P., & Iodice, M. (2024). THE LITTLE THINGS THAT RUN THE WORLD: Insetti a confronto attraverso modi di dire in latino, italiano, inglese [Insects compared through Latin, Italian, English idioms.]. *Lingue e Linguaggi*, 61.

Coulson, S. J., Gillespie, M. A., Loonen, M., Fossey, F., Sørensen, J. G., Gwiazdowicz, D. J., ... & Convey, P. (2025). Terrestrial and freshwater invertebrate science in Svalbard in a changing world: from regional to pan-Arctic scales (STAFIEN-Svalbard Terrestrial and Freshwater Invertebrate Ecology Network). *The State of Environmental Science in Svalbard—an annual report*.

[Two unspecified flea species are listed as in Svalbard.]

Da Silva, P. G., MacRae, T. C., Rank, N. E., Resh, V. H., Tauber, C. A., & Zuparko, R. L. (2024). 100 years of The Pan-Pacific Entomologist: A century of continuity and change. *The Pan-Pacific Entomologist*, 100(1), 1-23.

Hribar, Lawrence J. 2024. Hirsutism as a Conservation Measure: Biodiversity and Conservation Should Also Include Parasites. *Life: The Excitement of Biology* 12(1) 17-29. <https://blaypublishers.com/wp-content/uploads/2024/12/hribar.pdf>

Ishii, N., & Ishii, N. I. Evaluation of artistic inspiration by biodiversity in lyrics of Japanese popular songs. *osf.io*

Lee, J. S., Lee, D. J., Lim, J., Lee, H., Kim, J., Choi, M. J., ... & Roh, S. J. (2024). Establishment of an integrated island species inventory for insects (Arthropoda: Insecta) in Korea. *Journal of Asia-Pacific Biodiversity*, 17(2), 250-254.

Katherine Garcia, Sarah Orlofske. ECTOPARASITE COMMUNITIES IN BARRED OWLS (STRIX VARIA) USING NEW AND ARCHIVED MUSEUM COLLECTIONS. 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, 2024, Denver, Colorado, USA.

Roxanne M Gasperetti, Rianna J Taylor, Vasyl V Tkach, Tyler J Achatz, Jeffrey A Bell, Mike Kinsella, Sarah A Orlofske. PARASITES OF THE AMERICAN WOODCOCK (SCOLOPAX MINOR) FROM WISCONSIN, USA. 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, 2024, Denver, Colorado, USA.

Sarah A Orlofske. DIGITIZING A PARASITE COLLECTION AT A PRIMARILY UNDERGRADUATE INSTITUTION: CHALLENGES AND OPPORTUNITIES. 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, 2024, Denver, Colorado, USA.

Natural history collections are essential to numerous disciplines in biology and digitization increases access and research productivity. However, many smaller, specialized collections housed at universities present challenges for digitization efforts even though they may yield important discoveries. The Stephen J. Taft Parasitology Collection in the University of Wisconsin – Stevens Point Museum of Natural History was established in 2004 when Dr. Stephen (Steve) J. Taft retired after 33 years of teaching and research at UWSP. The mission of the Parasitology Collection is to serve as a foundation for research and training, primarily of undergraduates, in parasitology and related scientific and applied disciplines. The collection was largely acquired and amassed through the work of Steve Taft and the numerous students he mentored over the decades of his tenure at UWSP. In 2020 faculty-mentored student curation and digitization of the arthropod components of the collection began with the support of the National Science Foundation through the collaborative Terrestrial Parasite Tracker Project. Challenges associated with digitization at a primarily undergraduate institution include limited time from faculty mentors due to high teaching responsibilities, lack of other museum management staff, and the increased training demands and turnover of undergraduate students. Solutions to these challenges include clear and consistent training materials, standardized workflows, peer-mentorship, meticulous recordkeeping, and quality control processes. Helping students develop research ideas related to museum collections helps maintain engagement in the sometimes tedious and repetitive tasks, allows creativity, and builds their confidence through giving presentations. Over four years 21 students contributed to the digitization of 9280 specimens including high resolution slide images and label data to the Ecdysis data portal for use by data aggregators like iDigBio and ultimately the larger community of scientists and educators. This digitization effort has ignited an interest in the UWSP arthropod museum specimens and supported the development of new research efforts aimed at comparing modern collections with historical records including ectoparasites of Great-Horned Owls, Barred Owls, Northern Saw-Whet Owls and other birds of prey. These new projects demonstrate the opportunities for new research projects resulting from access to digital collections.

Marin Milotic, Alan J Lymbery, Andrew R C Thompson, Leighton J Thomas, Adria R A Eda, Stephanie S Godfrey. 2024. Parasite communities of critically endangered hosts are influenced by different sites of translocation as revealed by metabarcoding. 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, Denver, Colorado, USA.

Translocation is one of the most important tools for conserving endangered animals. However, conservation-driven translocation, aimed at helping the host species, often neglects symbiotic organisms such as parasites and may even lead to parasite losses, extirpation and extinction. Here, we examined nematode and coccidia parasites of translocated populations of the critically endangered woylie (*Bettongia penicillata*) found in four sanctuaries across Australia (Perup Sanctuary, Mt Gibson, Scotia and Yookamurra). Faecal samples of woylies were collected at each location in 2019. We aimed to compare the parasite communities of these four populations via metabarcoding of faecal samples using taxon-specific metabarcoding primers. While each location had a similar component community diversity, we found that woylies in different sanctuaries had distinct parasite infracommunities in terms of prevalence of infection, mean infracommunity diversity, and infracommunity composition. Contrary to our expectations, estimates of parasite prevalence and diversity were not strongly related to host density, geographic diversity of founding host population or time since establishment of the sanctuary. Instead, parasite prevalence and diversity appeared to be influenced more strongly by stochastic environmental factors during and after translocation, so that woylies in different translocation sites had distinctly different parasite communities despite considerable overlap in source populations. Such differences could potentially lead to displacement, extirpation or even extinction of certain rare or endemic parasites if they cannot be maintained in translocation sites, with repercussions for both the

health of the host and the wider ecosystem.

Salis, R., Sunde, J., Gubonin, N., Franzén, M., & Forsman, A. (2024). Performance of DNA metabarcoding, standard barcoding and morphological approaches in the identification of insect biodiversity. *Molecular Ecology Resources*, 24(8), e14018.

Salkeld, D., Skylar Hopkins, and David Hayman. 2024. *Emerging Zoonotic and Wildlife Pathogens: Disease Ecology, Epidemiology, and Conservation*. Oxford U. Press. ISBN: 9780198825920

Hosts and Distribution

Achyar, T. (2025). Ektoparasit pengganggu golongan filum arthropoda di lingkungan kandang semi terbuka (enclosure) orangutan sumatera (*Pongo abelii*) taman hewan pematang siantar sebagai referensi mata kuliah parasitologi (Doctoral dissertation, UIN Ar-Raniry Banda Aceh). [Ectoparasites of arthropod phylum in semi-open enclosure environment of Sumatran orangutan (*Pongo abelii*) in Pematang Siantar Zoo as a reference for parasitology course]

明尾亮佑, 石井秀空, 千葉駿, 栗山武夫, & 山崎健史. (2025). 兵庫県南部のアライグマから採集されたノミ類. *人と自然*, 35, 52-55. [Akio, R., Ishii, H., Chiba, S., Kuriyama, T., & Yamazaki, T. (2025). Fleas collected from raccoons in southern Hyogo Prefecture. *Human and Nature*, 35, 52-55.]

Bagheri, G., Hosseini-Vasoukolaei, N., Motevalli Haghi, F., Enayati, A., Fazeli Dinan, M., Nikookar, H., & Sahraee, F. (2024). Fauna and Biodiversity Investigations of Fleas Caught in Livestock Farms: Neka City, Mazandaran Province, Iran. *Journal of health research in community*, 10(3), 30-39.

Berihun, A. M., Yekoye, Y. D., & Alemayehu, M. M. (2024). Prevalence of domestic chicken ectoparasites in the east Goejam, Amhara, Ethiopia. *Veterinary Parasitology: Regional Studies and Reports*, 56, 101144.

Ourida Chebbah, Karim Souttou, Karim Ouachek, Mohamed Lounis, Sophie Brun et al. 2025. Inter- and Intra-Species Variation and Genetic Diversity of Flea Ectoparasites in Hedgehogs (Mammalia, Erinaceidae) Collected in Northern Algeria. *Insects* 2025, 16, doi: 10.3390/insects16040390

de Almeida, A. T. D. S., de Fátima Barbosa, N., & Alvarez, G. M. (2024). Conhecimento dos tutores sobre ectoparasitas em cães e gatos na região do Sul de Minas Gerais. *Pubvet*, 18(08), e1635-e1635. [Owners' knowledge about ectoparasites in dogs and cats in the southern region of Minas Gerais.]

Eren, G. (2024). New Data on Ectoparasites of the Caucasian Squirrel *Sciurus anomalus* (Rodentia: Sciuridae) in Türkiye: A Case Report. *Türkiye Parazitol Derg*, 48(4), 261-263.

Farmani, M., Spotin, A., Dolatkah, A., Gholizadeh, S., & Hazratian, T. (2024). Molecular Characterization and Phylogenetic Analysis of Flea Species in Human and Livestock Residence by Targeting ITS2 Region in East Azerbaijan Province, Iran. *Journal of Arthropod-Borne Diseases*.

Gharsan, F. N. (2024). Genetic identification of the sticktight flea *Echidnophaga gallinacea* (Westwood) infesting chickens in the Al-Baha region, Saudi Arabia. *Journal of Umm Al-Qura University for Applied Sciences*, 10(3), 485-490.

Ghasemi, F., & Miri, M. (2024). Exploring bat parasites in selected caves of Fars Province, Southern Iran. *Journal of Animal*.4.

Kara S. Gibson,Amber Born, Nancy C. Johnson,Deborah A. Neher,Anita J. Antoninka.2024. Too many mites, too little time: Expeditionous methods for manipulating and characterizing soil microarthropod communities. Entomological Society of America annual meeting, Nov.10-13, Phoenix, Arizona, USA.

Soil ecologists studying microarthropods are blessed (or plagued, depending on your perspective) with samples each containing hundreds to thousands of individuals belonging to tens to hundreds of morphotypes, of which most are typically mites. Unfortunately, time and research budgets are very rarely commensurate with the quantity and diversity of specimens collected. In this talk, we will review some strategies and tools that can help researchers to leverage limited resources when characterizing complex soil microarthropod communities. We will also present a low-cost, low-tech field mesocosm method for medium-term (1 to a few years) manipulation of soil and litter microarthropod assemblages by size. Units are constructed from 20 cm diameter PVC sewer pipe with large windows cut in the sides, wrapped in mesh of various sizes and capped with removable 40 µm mesh lids. These mesocosms were designed to reduce microclimate differences while also permitting repeated internal sampling of soil and litter, retrieval of standard substrates in decomposition studies, and gas flux analysis using LI-COR survey chambers. We tested this mesocosm method at two forest sites and found that mesh and defaunation worked effectively to manipulate the soil microarthropod community. Challenges and considerations of field mesocosms will be discussed, including microclimate differences and their measurement, defaunation methods and their side effects, and quantification of ecological functions performed by microarthropods. Together, this group of tools can enhance our ability to understand the functional roles of mites in ecosystems.

Graham B Goodman, Hannah Huston, Devon Adamo, Ruben F Navarro, Bradley J Cosentino 2024 Effect of coat color on parasitism in eastern gray squirrels . 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, Denver, Colorado, USA.

González-Rete, B., Jiménez-Cortés, J. G., Cabrera-Bravo, M., Salazar-Schettino, P. M., & Laura, A. (2024). Insect vectors of human pathogens in a warming world. *Effects of Climate Change on Insects: Physiological, Evolutionary, and Ecological Responses*, 287.

Gosik, R., Nekrutov, I., & Chobotow, J. (2024). Ectoparasites collected from dead bats (Chiroptera: Vespertilionidae) in east-central Poland. *Annals of Parasitology*, 70(3), 125-135.

Harimalala, M., Harimanana, R. R., Hamza, A. A., & Girod, R. (2024). Surveillance of Fleas and Their Small Mammal Hosts for Plague Risks in Some Main Seaports of the Islands of the Southwestern Indian Ocean. *The American Journal of Tropical Medicine and Hygiene*, 110(2), 311.

Hernández-Urbina, C. F., Fernández, J. A., Titulaer, M., Aguilar-Palma, N. G., Acosta, R., & Camargo-Sanabria, A. A. (2025). How much do we know about wild canid (Carnivora: Canidae) ectoparasites in Mexico? Current state of knowledge. *Mammalia*, (0).

Júnior, L. F. D., Cunha, I. N., Jorge, F. R., Graciolli, G., Bassini-Silva, R., Jacinavicius, F. C., ... & Camargo, L. M. A. (2024). Ectoparasitas em morcegos (Mammalia: Chiroptera) e seus patógenos na

região central de Rondônia, oeste da Amazônia brasileira.[Ectoparasites in bats (Mammalia: Chiroptera) and their pathogens in the central region of Rondônia, western Brazilian Amazon.] Pathogens, 14(4), 338.

Котти, Б. К., & Иванов, А. Л. (2024). Блохи (Siphonaptera) песчанок на Кавказе. Паразитология, 58(3), 240-251. Kotti, B. K., & Ivanova, A. L. (2024). Fleas (Siphonaptera) of Gerbils in the Caucasus. Parazitologîa, 58(3), 240-251.

Kotti, B. K., & Ivanov, A. L. (2024). Fleas (Siphonaptera) of Gerbils in the Caucasus. Entomological Review, 104(2), 129-137.

Kotti, B. K., & Ivanov, A. L. (2024). Fleas (Siphonaptera) of eulipotyphla in the Caucasus. Entomological Review, 104(7), 435-442.

Котти, Б. К., & Шкарлет, К. Ю. (2024). БЛОХИ (SIPHONAPTERA) В ЛЕСНОМ ПОЯСЕ ЗАПАДНОГО КАВКАЗА. ISSN 3034-2392 ВЫПУСК 20, 172. [Cotti, B. K., & Shkarlet, K. Y. (2024). FLEAS (SIPHONAPTERA) IN THE FOREST BELT OF THE WESTERN CAUCASUS. ISSN 3034-2392 ISSUE 20, 172.]

Kotti, B. K. (2024). Fleas (Siphonaptera) of Rodents in the Transcaucasian Highland Natural Plague Focus. Entomological Review, 104(4), 259-266.

The Transcaucasian highland vole-based natural plague focus is located on the Javakheti and Armenian plateaus and the Lesser Caucasus. The rodents distributed in this region are hosts to 42 species of fleas from 21 genera. One-third of the flea species are monoxenous parasites, and the remaining species are oligoxenous or pleioxenous parasites. The vole parasite *Nosopsyllus consimilis* is typical of the foothills and the middle mountain belt, while such vole parasites as *Amalaraeus dissimilis daghestanicus*, *Callopsylla c. caspia*, *C. s. saxatilis*, *Paradoxopsyllus h. hesperius*, *Frontopsylla c. caucasica*, *Amphipsylla kuznetzovi*, *Rhadinopsylla caucasica*, and *Paraneopsylla dampfi* occur in the highlands. The ranges of 24 flea species cover the entire plague focus, while other species are confined to the Armenian Plateau (*Citellophilus transcausicus*, *Oropsylla idahoensis ilovaiskii*, *Neopsylla setosa spinea*, *Ctenophthalmus iranensis*, *C. shovi*, *C. secundus*), the Javakheti Plateau (*Amphipsylla georgica*), or the Lesser Caucasus (*C. wladimiri*). The warm season is the most favorable time for plague transmission in the Transcaucasian focus, since fleas infesting the common vole actively feed and reproduce during this period. The most active pathogen vector among them is *Callopsylla c. caspia*.

Кондратьев, Е. Н. ПРОСТРАНСТВЕННОЕ РАСПРЕДЕЛЕНИЕ ЧЛЕНИСТОНОГИХ В НОРАХ БЕРЕГОВОЙ ЛЯСТОЧКИ (RIPARIA RIPARIA (LINNAEUS, 1758)) НА СЕВЕРЕ НИЖНЕГО ПОВОЛЖЬЯ. [Kondratiev, E. N. SPATIAL DISTRIBUTION OF ARTHROPODS IN THE BURROWS OF THE SAND martin (RIPARIA RIPARIA (LINNAEUS, 1758)) IN THE NORTH OF THE LOWER VOLGA REGION.] <https://phsreda.com/e-articles/10586/Action10586-110164.pdf>

[*Ceratophyllus styx*]

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(Insecta, Siphonaptera) of the Republic of Karelia. *Entomological Review*, 104(6), 407-420.

This paper explores the specific features of the fauna of fleas (Siphonaptera) of Karelia, as part of a larger geographic region comprising Northwestern Russia and Fennoscandia. Our analysis included information from the available summary reviews and data from the PARHOST1 information analysis system on the world flea fauna, hosted by the Zoological Institute of the Russian Academy of Sciences and supplemented regularly since 2001. At present, 28 flea species of 13 genera have been recorded for Karelia, amounting to about 35% of the regional fauna and 13% of the flea fauna of Russia. The species occurring in Northwestern Russia but not found in Karelia can be divided into two major groups. The first group includes the parasites of those hosts whose fleas remain unstudied. The second group comprises the flea species that are absent in the Karelian fauna due to vicariation.

Keywords: parasites, fleas, species diversity, rodents, insectivores, Northwestern Russia, Fennoscandia

Minaya, D., Flores-Bancayan, C., Pellón, J. J., Iannaccone, J., & Lareschi, M. (2024). Fleas (Insecta: Siphonaptera) of Peru: A Review of Their Diversity and the Host Specificity in the Country. *Diversity*, 16(12), 762. <https://doi.org/10.3390/d16120762>

In the Neotropical biogeographic region, there are about 290 known species and subspecies of fleas, which have as their main hosts to the rodent subfamily Sigmodontinae. In Peru, more than 22 years have passed since the last list of known flea species was published. This study presents a review and update of the fleas recorded in Peru from 1908 to 2024. This list contains information on the flea species, including their known hosts, the localities in Peru where they were reported, the main references and some comments. In addition, the specificity of flea–host interactions was assessed using an interaction network approach. In the literature review, 86 flea species were found distributed among 9 families, 34 genera and 19 subspecies. These fleas were recorded parasitizing at least 71 species of mammals and birds (14 orders, 28 families, 54 genera), 15 reported only at the genera level and seven undetermined ones. The flea–host interaction network exhibited a modular structure, indicating that these interactions tend to be specialized. A sigmodontine rodent (*Akodon mollis*) and a Rhopalopsyllidae flea (*Polygenis litargus*) were highly connected across modules and the overall network, and thus were important to maintain the structure. This review contributes to the knowledge of fleas in Peru, synthesizing the scattered information and providing information on the interaction they have with their hosts.

Muñoz-Saba, Y., Neita-Moreno, J. C., Perdomo-Gaitán, M. A., Gómez-Hernández, J., Sauro, F., & Lasso, C. A. (2024). Biota asociada al guano de murciélagos en cuevas cuarzoareníticas de la Amazonia colombiana. *Boletín Geológico*, 51(2). [Biota associated with bat guano in quartz-sand caves of the Colombian Amazon.]

中村有加里, ナカムラユカリ, 深瀬徹, & フカセトオル. (2024). Infection of the House Mouse *Mus musculus* (Rodentia: Muridae) Captured in Zushi-shi, Kanagawa Prefecture, Japan with the Cat Flea *Ctenocephalides felis* (Siphonaptera: Pulicidae). [Infection of the House Mouse *Mus musculus* (Rodentia: Muridae) Captured in Zushi-shi, Kanagawa Prefecture, Japan with the Cat Flea *Ctenocephalides felis* (Siphonaptera: Pulicidae). Yukari NAKAMURA* and Tohru FUKASE] 岡山理科大学紀要. A, 自然科学, 60, 261-264. [THE BULLETIN OF OKAYAMA UNIVERSITY OF

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Corea, D. A. N., Flores, J. L., Baak, C. M. B., Fuentes, Y. M. O., Chávez, E. C., & Rejón, J. E. G. (2024). Ectosimbiontes de roedores sinantrópicos de Celestún, Yucatán, México. Nuevos registros de ácaros mióbidos y miocóptidos para la península de Yucatán. ACTA ZOOLOGICA MEXICANA (NS), 1-15. [Ectosymbionts of synanthropic rodents from Celestún, Yucatán, Mexico. New records of myobid and myocoptid mites from the Yucatán Peninsula.]

Núñez-Corea, D. A., Loría-Cervera, E. N., Sosa-Bibiano, E. I., López-Ávila, K. B., Baak-Baak, C. M., Cab-Cauich, I. Y., ... & García-Rejón, J. E. (2024). Hematophagous Ectoparasites of Wild and Synanthropic Rodents in Yucatan, Mexico. Southwestern Entomologist, 49(4), 1-11.

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Josh Pletcher, Cassie Stitzman, Kurt Galbreath. 2024. Flea fauna of Mongolian pikas, jerboas, and zokors. 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, Denver, Colorado, USA.

Despite their medical and veterinary importance, knowledge of the distribution and host associations of most flea species is poor. Documenting the biogeography and ecology of fleas can improve our understanding of host dispersal, disease transmission dynamics, and environmental change. Recent site-intensive surveys of small mammals and their parasites in Mongolia provide an opportunity to investigate patterns of flea diversity across ecosystems and mammalian hosts. Mongolia lies at a major zone of ecological transition from the xeric Gobi Desert in the south through steppe biomes to boreal forests in the north. Documenting the complexity of mammal/flea interactions and their global distributions supports future systematic, ecological, and epidemiological studies, and provides an important baseline for understanding the impact of ecological disruption in a region that is already experiencing severe effects of ongoing climate change. To explore how flea community composition changes with geography and host, we are characterizing diversity and abundance of flea species associated with Mongolian pikas, jerboas, and zokors. These small mammals inhabit desert, steppe, and alpine environments, and include fossorial, semi-fossorial, and non-burrowing species. Some of these species form large colonies, while others are solitary. We extracted DNA from representative flea species in preparation for future genomic sequencing, and cleared their exoskeletons for morphological identification using published keys. Fleas identified from the burrowing pikas *Ochotona dauurica* and *Ochotona pallasi* and the rock-dwelling pika *Ochotona hyperborea* include genera from the families Pulicidae, Hystrihopsyllidae, Leptopsyllidae, and Ceratophyllidae. Five out of twelve genera identified are primarily associated with pikas: *Amphalius*, *Callopsylla*, *Ctenophyllus*, *Ochotonobius*, and *Paramonopsyllus*. The genera *Amphalius* and *Ctenophyllus* were common to all sampled pika species. Other genera (*Amphipsylla*, *Echidnophaga*, *Frontopsylla*, *Paradoxopsyllus*, *Peromyscopsylla*, *Rhadinopsylla*) recovered are host-generalists or are primarily associated with

rodents occupying similar habitats. *Echidnophaga*, *Paradoxopsyllus*, and *Paramonopsyllus* were only recorded on *O. pallasi* in far western Mongolia. *Ochotona dauurica* and *O. hyperborea* occurred in close proximity near Lake Khuvsgul, and shared more flea genera (*Amphalius*, *Ctenophyllus*, *Frontopsylla*, *Rhadinopsylla*) than expected due to their different life histories. Many fleas are specialized for particular habitats, but ecological disruptions can facilitate exchange of fleas between typically isolated host species.

Romashova, E. B., Medvedev, S. G., & Kuznetsov, Y. E. (2024). Distribution of the flea *Monopsyllus Sciurorum Sciurorum* (Schrank, 1803)(Siphonaptera, Ceratophyllidae) in the wild and in fur farms. *Parazitologîa*, 58(4), 338-350. Romashova, E. B., Medvedev, S. G., & Kuznetsov, Y. E. (2024). Distribution of the flea *Monopsyllus Sciurorum Sciurorum* (Schrank, 1803)(Siphonaptera, Ceratophyllidae) in the wild and in fur farms. *Entomological Review*, 104(3), 214-221.

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Quelle & Meyer Verlag GmbH & Co..[Brohmer–The fauna of Germany and neighboring countries.]

Seidi, S., Tavassoli, M., Malekifard, F., & Ben Said, M. (2024). Molecular identification, multigenic characterization and phylogenetic analysis of *Ctenocephalides canis* fleas from humans and domestic animals in Iran. *Acta Parasitologica*, 69(2), 1219-1230.

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Thoroughgood-Smith, J. J. (2025). DIVERSITY AND ABUNDANCE OF FLEAS ON RICHARDSON'S GROUND SQUIRRELS AND IN THEIR BURROWS, AND THE BACTERIA WITHIN THE MICROBIOMES OF THESE FLEAS (Doctoral dissertation).

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Виноградова, Е. А. (2024). ФАУНА ЭКТОПАРАЗИТОВ НОЧНИЦЫ БРАНДТА MYOTIS BRANDTII (EVERSMANN, 1845), ОБИТАЮЩЕЙ НА ТЕРРИТОРИИ ТВЕРСКОЙ ОБЛАСТИ. Ответственный редактор, 9. [Vinogradova, E. A. (2024). FAUNA OF ECTOPARASITES OF BRANDT'S BAT MYOTIS BRANDTII (EVERSMANN, 1845), LIVING IN THE TERRITORY OF TVER REGION. Editor-in-Chief, 9.]

Wang, J. S., & Jia, L. J. (2024). First discovery of snow scorpionflies (Mecoptera: Boreidae) from China. *The Indochina Entomologist*, 1(11), 75-93.

Yusupova, A. (2025). THE ENTOMOFAUNA OF THE LOW HILLS OF KARAKALPAKSTAN. Академические исследования в современной науке, 4(7), 119-125.

Morphology

Calandriello, D. C., Cunha, V. A., Batista, D., & Genevcus, B. C. (2025). Genetic architecture of morphological adaptation and plasticity in insects: gaps, biases, and future directions. *Current Opinion in Insect Science*, 101362.

Insects exhibit a vast array of morphological specializations. Recent eco-evo-devo studies have provided a fresh perspective into how insect morphology can respond to the environment, both plastically and adaptively. Here, we performed a systematic literature analysis to identify biases and gaps in research on the molecular mechanisms underlying insect morphological adaptation and plasticity. We found that plasticity studies are increasingly present in the literature, while adaptation studies lag behind. Additionally, we observed disproportionate focus on a few insect orders and specific traits like wings and body size. We highlight the need to explore the broader insect diversity, including understudied groups and unexplored traits like reproductive organs, as well as utilize advanced methods to capture subtle morphological variation. Studying a wider range of species with diverse morphologies and ecological features, as well as implementing modern genome-wide tools, can reveal the full spectrum of mechanisms underlying morphological adaptation and plasticity in insects.

Lyu, Q. H., Xue, S., Wei, X. F., & Dong, W. J. (2024). Ultrastructure of the ejaculatory duct of *Terrobittacus implicatus* (Mecoptera: Bittacidae). *Protoplasma*, 1-8.

Petrov, P. N., & Polilov, A. A. (2025). Ecomorphology of microinsects. In *Insect Ecomorphology* (pp. 439-456). Academic Press.

Miniaturization is a major trend of insect evolution. At least nine insect orders include microinsect species (less than 2 mm long). Representatives of many lineages have independently evolved into microinsects, the ecology of which is quite diverse. Some of them are strongly modified, and some are rather similar morphologically to much larger related species. Paradoxically, while smaller animals tend to be r-strategists, extremely small insects tend to be K-strategists, because of limitations to the minimum size of a viable egg. The minimum size of a nervous system providing for their relatively complex behaviours probably also limits the minimum body size of insects. In extreme cases, insect miniaturization leads to the loss of some organs and functions, but the vast majority of microinsects retain all such functions. Further studies of the ecomorphology of microinsects should include more species and a broader array of methods. Microinsects can serve as useful models for ecomorphology.

[Editor's comment: *Tunga penetrans* (sand flea, jigger, chigre, chigoe flea) is likely the smallest adult flea at 1 mm long (before hypertrophic neosomy swells females up). Miniaturization seems to have evolved several different times in "nest fleas", too.

Tunga is small because it does not rely on the host having a nest, it relies upon the embedded female flea, whereas a nest flea is small because the nest is well-developed and can be relied upon for the immatures' food and shelter.

Are small fleas facing challenges of small eggs, minimal nervous systems, and loss of organs and functions? Are small fleas K-strategists?]

Physiology

Allman, D. M. (2024). The jumps of Australian Myrmecia ants (Doctoral dissertation, Macquarie University).

Cooke, M., Chembars, M. S., & Pitts, R. J. (2025). The conserved IR75 subfamily mediates carboxylic acid detection in insects of public health and agricultural importance. *Journal of Insect Science*, 25(1), ieaf012.

Insects perceive and respond to carboxylic acids (CAs), amines, and aldehydes primarily via conserved ionotropic receptors (IRs). These receptors form the basis for a second olfactory system distinct from the well-characterized odorant receptors. Neurons expressing IRs are housed in dedicated sensilla and innervate glomeruli, separate from those innervated by odorant receptor neurons. The IR8a co-receptor is highly conserved across insect orders and, together with ionotropic receptor tuning receptors, primarily detects CAs. The conservation of genes and the anatomical separation of neural pathways underscore the importance of these compound classes and CAs, specifically in insect chemical ecology. We provide a summary of carboxylic acid detection in insects, focusing on dipteran and lepidopteran species of significance to public health and agriculture. An overview of insect

behavior toward CAs is provided, as well as a comprehensive update on carboxylic acid receptor function in insects. Phylogenetic analysis of publicly available genome databases reveals several species that encode and express homologs of previously orphanized carboxylic acid receptors, highlighting avenues for future research.

Keyes-Scott, N. I. (2024). Factors that Modulate Insect Developmental, Metabolic and Reproductive Physiology in Two Vectors of Human and Animal Disease (Doctoral dissertation, University of Georgia).

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Katie A. Weber, 2024. Transcriptional Induction of Immune Pathway Genes and Subsequent Antibacterial Activity in the Digestive Tract of the Cat Flea (*Ctenocephalides felis*). Master of Science in Biology (M.S.) Thesis. Georgia Southern University.

Zhang, B., Chen, L., Xiao, S., Dang, C., Wang, F., Fang, Q., Ye, X., Stanley, D.W. and Ye, G. (2024), iSalivaomicDB: A comprehensive saliva omics database for insects. *Insect Science*, 31: 1998-2001. <https://doi.org/10.1111/1744-7917.13349>

Evolution

Chen, B., Duan, M., Liu, S., Liu, Y., Tang, S., Jiang, D., ... & Yang, X. (2024). The complete mitochondrial genome and phylogenetic implications of *Paradoxopsyllus custodis* and *Stenischia*

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Mecoptera is a small relict order of insects within the Holometabola. Panorpidae is the most speciose family in Mecoptera. They are also known as scorpion flies due to the enlarged and upward recurved male genital bulb. *Panorpa liui* Hua, 1997, a member of Panorpidae, is a bivoltine species of seasonal color polyphenism in the lowland plain of northeastern China. In this study, we applied PacBio HiFi and Hi-C sequencing technologies to generate a chromosome-level genome reference of *P. liui*. The assembled genome is 678.26 Mbp in size, with 91.3% being anchored onto 23 pseudo-chromosomes. Benchmarking Universal Single-Copy Orthologs (BUSCO) estimation reveals the completeness of this assembly as 95.1%. By integrating full-length transcriptome and homologs of related species, we generated full annotation of this assembly, yielding a total of 15,960 protein-coding genes, of these, 15,892 genes were anchored on the 23 chromosomes. The high-quality genome provides critical genomic resources for population genetics and phylogenetic research on Mecoptera. It also offers valuable information for exploring the mechanisms underlying seasonal color polymorphism.

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Košša, J., Ambros, M., Dudich, A., & Baláž, I. (2024). Impact of environmental factors on small mammals and their fleas. *Folia Oecologica*, 51(2), 146-153.

Small mammals in the Horšianska dolina-Valley were researched first in 1984–86 and later in 2016–19. This paper seeks to identify and assess the species composition and dynamics of fleas (Siphonaptera) found in the valley on small mammals of Rodentia and Eulipotyphla and to compare findings from these two periods separated by thirty years. The results indicated deviations in small mammal and flea community similarity as well as in the small mammal indicator species for each habitat and flea indicator species for the small mammal species. A total 12 different species of small mammals were captured, and 12 flea species were collected. Eight flea species were detected in 1984–86 and ten species were found in 2016–19. In both periods *A. penicilliger*, *C. assimilis*, *C. solutus*, *D. dasyncema*, *N. fasciatus* were captured. The differences were that in the earlier period, the species uncovered were *C. congener*, *P. bidentata*, and *P. soricis*, while in the later period they were *C. agyrtes*, *C. bisoctodontatus*, *M. turbidus*, and *R. integella*. In the relationship between fleas and small mammals, *C. solutus* was the specific species for *Apodemus flavicollis*, while *A. penicilliger* and *C. agyrtes* were for *Clethrionomys glareolus*, and *C. assimilis* was for *Microtus arvalis*. Keywords Eulipotyphla, indicator species, Rodentia, Siphonaptera

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Ticks are important vectors and reservoirs of pathogens causing zoonotic diseases in camels and other livestock, rodents and other small mammals, birds and humans. *Hyalomma dromedarii* is the most abundant tick species in Saudi Arabia and United Arab Emirates (UAE) affecting primarily camels, and to a lesser extent, other livestock. Species presence data, land use/landcover, elevation, slope and 19 bioclimatic variables were used to model current and future distribution of *H. dromedarii* ticks using maximum entropy species distribution modelling (MaxEnt.). The model highlighted areas in the northern, eastern and southwestern parts of the study area as highly suitable for ticks. Several variables including land use/land cover (LULC) (53.1%), precipitation of coldest quarter (Bio19)

(21.8%), elevation (20.6%), isothermality (Bio3) (1.9%), mean diurnal range [mean of monthly (max temp – min temp)] (Bio2) (1.8%), slope (0.5%), precipitation, seasonality (Bio15) (0.2%) influenced habitat suitability of ticks, predicting high tick density or abundance. Middle of the road scenario (ssp2-4.5) where CO₂ levels remain similar to current levels, did not indicate a major change in the tick distributions. This tick distribution model could be used for targeting surveillance efforts and increasing the efficiency and accuracy of public health investigations and vector control strategies.

Ramírez-Chaves, H. E., Ospina-Pérez, E. M., Ossa-López, P. A., & Rivera-Páez, F. A. Diversity and Interactions between Ectoparasites Associated with Mammals (Mammalia) in the Department of Arauca, Orinoquia Region of Colombia. Orinoquia Region of Colombia. papers.ssrn.com

Anthony Raona, Michael Joyce, Joseph Jacquot, Paul Keenlance. 2024. Parasite Mediated Competition between Flying Squirrels (*Glaucomys*). 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, 2024, Denver, Colorado, USA.

Southern flying squirrels (*Glaucomys volans*) are rapidly expanding their range north due to warming temperatures. This range expansion has been implicated in a concurrent reduction in northern flying squirrel (*Glaucomys sabrinus*) range. The reduction in range has also led to conservation concern for northern flying squirrels throughout much of the Great Lakes Region prompting research interest in flying squirrel interactions. The nematode *Strongyloides robustus* is an intestinal parasite infecting most sciurid species. Previous research suggests that *S. robustus* severely worsens northern flying squirrel health while minimally affecting southern flying squirrels. Thus, interactions between the two species are influenced at least in part through parasite mediated competition. *G. volans* have traditionally been believed to be the primary reservoir for *S. robustus* in areas of sympatry, with colonization exposing resident *G. sabrinus* populations to the parasite and resultant reductions in *G. sabrinus* populations. Recent research has suggested that sciurid species other than *G. volans* may be serving as reservoirs for the parasite which would require a major rethinking of our understanding of the interactions between the two species. Squirrels will be tested for *S. robustus* via a modified McMaster fecal float technique, from which parasite eggs are identifiable in the feces. Squirrel trapping will take place through 2023 and 2024. This research aims to determine the prevalence of *S. robustus* among sciurid species in areas where both northern and southern flying squirrels are present in northern Minnesota, and where only northern flying squirrels are present in northern Minnesota. This project is still a work in progress, but only one positive sample of 94 suggests that *S. robustus* and parasite mediated competition do not play a significant role in this system. The conclusions we draw will further inform the ecology of how northern and southern flying squirrels indirectly interact between themselves, other sciurids, and a parasite of concern.

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Parasite species interactions, host biology traits, and external environmental factors can drive co-occurrence patterns between parasite species. We investigated co-occurrence patterns between three ectoparasite species (mite (*Neotrombicula harperi*), and fleas (*Orchopeas caedens* and *Ceratophyllus vison*)) of North American red squirrels (*Tamiasciurus hudsonicus*). We evaluated (1) whether ectoparasites of red squirrels exhibit non-random co-occurrence patterns, and (2) the contribution of host and external environmental factors to parasite co-occurrence. Bayesian ordination and regression analysis (boral) revealed random associations between parasite species pairs when accounting for host and external environmental factors. However, the mite *N. harperi* exhibited a negative association with the flea *O. caedens* and positive association with the flea *C. vison* linked to temporal patterns of occurrence. Our data suggests that parasites of the investigated population of red squirrels tend to form associations based on temporal trends in infestation rather than species interactions. Further experimentation should investigate the role of additional factors on parasite co-occurrence patterns, such as temperature, precipitation, and humidity.

Yan, H., Bu, F., Wu, X., Zhang, H., Sun, S., Bai, W., ... & Fu, H. (2024). Host selection and influencing factors of parasitic fleas on the body surface of desert rodents, Inner Mongolia, China. *International Journal for Parasitology: Parasites and Wildlife*, 25, 100993.

Fleas frequently parasitize rodents and serve as major vectors for many zoonotic diseases. The Alxa Desert in Inner Mongolia, China, is a typical arid region in Asia, which is suitable for the coexistence of a variety of rodents in this environment. Understanding the host selection of fleas among these rodents and the factors influencing this selection is crucial for comprehending rodent-borne diseases and the vector roles of fleas. This study utilized live-trap methods to capture rodents coexisting in the Alxa Desert from April to October 2022. Body surface fleas were collected and analyzed for abundance across different hosts, sexes, and reproductive statuses. The results indicated that: (1) there were significant differences in the species and abundance of parasitic fleas on different rodent species, with *Ophthalmopsylla kiritschenkoi* and *Mesopsylla hebes* preferring the *Dipus sagitta* and *Orientallactaga sibirica* as their hosts, and *Xenopsylla conformis* preferring *Meriones meridianus* as its host. (2) Fleas exhibited sex-preferential host selection, showing a greater preference for male hosts. (3) Among rodents of different reproductive states, most fleas prefer to parasitize individuals in non-reproductive periods. (4) The mean abundance of fleas varied significantly between seasons ($P < 0.05$), with spring and autumn being the peak periods for flea parasitism. (5) Factors influencing the mean abundance of fleas included host species, sex, reproductive status, and season, with host species having the greatest impact.

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The distribution and host-parasite relationships of fleas of the genus *Xenopsylla* (Pulicidae) and their significance as plague vectors are analyzed. The worldwide distribution of plague foci and the role of this flea genus in pathogen circulation in various regions are considered. *Xenopsylla* fleas either act as the only main vectors or are included in the main vector pool in 114 out of 255 known natural plague foci around the world. Spontaneous infection with the plague microbe has been detected for 21 species of this genus, and 17 species are considered the main plague vectors in some foci. The flea *X. cheopis* is the most significant plague vector in Africa and South-East Asia.

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The Oriental rat flea, *Xenopsylla cheopis*, is known worldwide as an efficient plague vector, including in Madagascar, where the disease remains a public health concern. Chemical control is the primary response method against *X. cheopis* in Madagascar. Previous bioassays focusing on different flea populations from Madagascar showed phenotypic resistance to various insecticides, including deltamethrin and fenitrothion, which, respectively, represent the previous and current chemicals for flea vector control. Despite apparent insecticide resistance, the associated mechanisms of this resistance remain poorly known. The aims of this study were to adjust diagnostic doses of deltamethrin and fenitrothion and to investigate the metabolism-based insecticide resistance of *X. cheopis* in Madagascar. Five available laboratory-reared flea strains of *X. cheopis* were selected, and their susceptibility statuses to deltamethrin and fenitrothion were determined using the WHO standard bioassay. Diagnostic doses of each insecticide were determined by the probit method, in accordance with concentration gradients. Biochemical microplate-based assays were performed to detect overproduction of cytochrome P450, alpha-/beta-esterases, and glutathione S-transferase (GST), which are signatures of metabolic resistance. The five tested strains showed different susceptibility statuses against deltamethrin and fenitrothion. The diagnostic doses were estimated to be 0.07% for deltamethrin and 1.56% for fenitrothion. Increased activities of cytochrome P450, beta-esterase, and GST enzymes in the resistant strains were revealed in comparison with those of the susceptible strain. In conclusion, readjusted diagnostic doses will help to better understand the susceptibility status of *X. cheopis* to deltamethrin and fenitrothion. The overproduction of cytochrome P450, beta-esterase, and GST observed on deltamethrin - resistant flea strains suggests metabolic resistance.

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Los sifonápteros, conocidos comúnmente como pulgas (Siphonaptera), constituyen un problema significativo de salud pública en varias regiones del mundo. En la región montañosa de Ecuador, se ha reportado un aumento en la frecuencia de casos de alergia a Siphonaptera en los últimos años, lo que destaca la urgencia de realizar investigaciones en este tema. **Objetivo.** Este estudio busca analizar la prevalencia de alergias a Siphonaptera en la región serrana de Ecuador, identificar los factores que contribuyen a su aparición y evaluar las estrategias de manejo y prevención existentes. **Metodología.** Esta investigación adopta un enfoque mixto, integrando métodos cuantitativos y cualitativos, con un alcance descriptivo y un diseño no experimental. Los estudios revisados se basan principalmente en técnicas de observación directa, pruebas clínicas y pruebas de laboratorio, empleadas para analizar los casos de alergia por Siphonaptera y sus posibles alérgenos. Se realizó una revisión sistemática y

analítica de los casos reportados en la región sierra, mismos que fueron comparados con datos de países europeos, y se abarcó un total de 28 artículos relacionados con la alergia a Siphonaptera. Resultados. La alergia a Siphonaptera y sus alérgenos se consideran un problema de salud pública relevante, dado que se ha documentado una alta prevalencia de infecciones asociadas. Estudios previos, como los de Cortez (2018) y Torrez (2023), han investigado protozoos y alérgenos relacionados. Se estima que aproximadamente el 60% de los animales, principalmente perros y gatos, en la región sierra de Ecuador, sufren algún tipo de alergia causada por pulgas y subespecies locales. Conclusiones. A partir de la revisión, se concluye que no existe un estudio exhaustivo sobre la alergia a Siphonaptera en la región, lo que sugiere la necesidad de un enfoque investigativo renovado sobre las infestaciones en diversas áreas de Ecuador.

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An Xie, Yiyue Zhang, Martin F. Breed, Xinli An, Haifeng Yao, Qiansheng Huang, Jianqiang Su, Xin Sun (2024) Terrestrial invertebrate hosts of human pathogens in urban ecosystems. *Eco-Environment & Health* 3 : 9–380..

Terrestrial invertebrates in urban ecosystems are extremely species-rich, have many important roles in material flow and energy circulation, and are host to many human pathogens that pose threats to human health. These invertebrates are widely distributed in urban areas, including both out- and indoor environments. Consequently, humans are frequently in contact with them, which provides many opportunities for them to pose human health risks. However, comprehensive knowledge on human pathogen transfer via invertebrates is lacking, with research to date primarily focused on dipterans (e.g., mosquitoes, flies). Here, we take a broad taxonomic approach and review terrestrial invertebrate hosts (incl. mosquitoes, flies, termites, cockroaches, mites, ticks, earthworms, collembola, fleas, snails, and beetles) of human pathogens, with a focus on transmission pathways. We also discuss how urbanization and global warming are likely to influence the communities of invertebrate hosts and have flow-on risks to human health. Finally, we identify current research gaps and provide perspectives on future directions.

Microbial Symbiontes

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Emily Greenman, Josh Sharp, Kurt Galbreath. BARTONELLA DIVERSITY IN SMALL MAMMALS OF MONTANA AND IDAHO. 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, 2024, Denver, Colorado, USA.

Hovorková, J. (2024). Blechy jako hostitelé a přenašeči bakterií. [Fleas as hosts and carriers of bacteria.] Bachelor's thesis, supervised by Hammerbauerová, Iva. Prague: Charles University, Faculty of Science, Department of Parasitology, 2024.

Fleas (Siphonaptera) are worldwide significant ectoparasites and vectors of several human pathogens. The most serious flea-born infections include plague, rickettsiosis and bartonellosis caused by *Yersinia pestis*, *Rickettsia* spp. and *Bartonella* spp. I have focused on the microbiome of fleas with an emphasis on bacteria which are transmitted by fleas or are pathogenic, mutualistic or commensal in relation to their flea host, from the point of view of their mutual interactions. The composition of the flea microbiome can be influenced both by mutual interactions between individual bacteria and by many different factors, which include, for example, the species or gender of flea or its host, humidity and temperature of the environment or geographic location.

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Sarah Mays Maestas, 2024. Vector and pathogen surveillance in south Texas: Understudied vector species Entomological Society of America annual meeting, Nov.10-13, Phoenix, Arizona, USA.

The University of Texas Rio Grande Valley is a minority-serving institution located in the south Texas trans-border region. The unique south Texas environments support vector species that are found in few other areas of the United States. The effects of changing climate on vector distribution, along with the recognition of many emerging and re-emerging vector-borne pathogens in recent years reflect the need for vector ecology research and improved vector management strategies. Current work in the recently-established Vector-borne Disease Ecology Laboratory focuses on better understanding vector, host, and pathogen interactions that contribute to vector-borne disease cycles in south Texas. Many native vector species in south Texas remain understudied, and little is known about the roles of many in the maintenance and transmission of vector-borne pathogens. South Texas is a hotspot for diseases such as tick-borne relapsing fever and murine typhus, but these disease systems require further study. Efforts are underway to screen tick and flea species for Rickettsial pathogens. Collaborative investigations of vector-host interactions have resulted in new host detections for the javelina flea, including the first reported detection of fleas on nilgai. Our ongoing soft tick research suggests the presence of cryptic species or species complexes that may be associated with human pathogens. Efforts are underway to collect and screen tick samples for tick-borne pathogens such as Hepatozoon and relapsing fever group *Borrelia*. These studies will help to better understand vector-borne disease dynamics and improve disease prevention efforts for human, companion animal, livestock, and wildlife health.

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Jordan Salomon, Emily Leeke, Haydee Montemayor, Cassandra Durden, Lisa Auckland, Sujata Balasubramanian, Gabriel L. Hamer., Sarah A. Hamer. 2024. D3025: On-host flea phenology and flea-borne pathogen surveillance among mammalian wildlife of the pineywoods of East Texas. Entomological Society of America annual meeting, Nov.10-13, Phoenix, Arizona, USA.

Flea-borne diseases are endemic in Texas, USA, with an increasing incidence of human disease including flea-borne typhus and cat scratch disease. Knowledge of flea natural history could provide critical information to protect public health, yet many knowledge gaps remain outside of plague-endemic regions. Our objective was to characterize seasonal activity patterns of fleas on common mammalian wildlife species and test fleas and wildlife for pathogens in the *Rickettsia* and *Bartonella* genera. We performed one year of monthly trapping for rodents and medium-sized mammals in a

national forest with high recreational use and urban encroachment in east Texas. Out of 90 mammal captures representing seven species, 101 fleas were collected representing *Polygenis* spp., *Ctenocephalides felis*, and *Orchopeas* species. Virginia opossums (*Didelphis virginianus*) hosted 99% of the collected fleas (100 fleas) and a single flea was found on an eastern woodrat (*Neotoma floridana*). Flea infestation prevalence of opossums was 79% (23/29). Mean flea abundance was 4.39 fleas, with fleas encountered across all seasons and intensity peaking in spring. One cat flea removed from an opossum was positive for *Bartonella henselae*. Furthermore, we identified tissue or blood of three raccoons (*Procyon lotor*) and one golden mouse (*Ochrotomys nuttalli*) positive for *Rickettsia amblyommatis*. Given the increasing public health concern of flea-borne disease, these findings provide an ecological basis for understanding the natural maintenance of vectors and pathogens from a sylvatic setting.

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[no fleas were reported as collected]

Cassie M Stitzman, Kurt E Galbreath. INVESTIGATING BARTONELLA PREVALENCE OF INTERMOUNTAIN WEST FLEAS . 99th annual meeting of the American Society of Parasitologists (ASP). Program, June 14-17, 2024, Denver, Colorado, USA.

Fleas are obligate parasites that can transmit blood-borne pathogens in wildlife populations, including small mammal communities in the northern Intermountain West. Some fleas have been documented to prefer specific hosts, whereas other fleas are associated with a broader range of hosts. These alternative patterns of host preference, along with the distribution of flea species across elevation gradients, may have implications for the distribution of vector-transmitted pathogens, such as *Bartonella* species, which are blood-borne, gram-negative bacteria. An understanding of these mammal/flea/*Bartonella* relationships may provide insight into potential pathogen spill-over as climate change continues to disrupt ecological communities. I am studying communities of fleas from 25 small mammal species collected from 18 montane localities in Idaho and Montana to better understand the relationship between pathogen diversity, flea diversity, and host specificity. I am extracting whole genomic DNA from 100 fleas, while retaining intact exoskeletons for species identification. To determine the prevalence of *Bartonella* among the fleas, I am using digital droplet PCR, an ultra-sensitive method for detecting low copy number DNA molecules in a sample. To assess the identity of *Bartonella* in positive samples, I will perform standard PCR to amplify and sequence the *gltA* gene. Sequences will be compared to known reference samples from GenBank using BLAST searches. The collected data will then undergo phylogenetic analysis with maximum likelihood methods. Preliminary results suggest that *Bartonella* is prevalent in diverse flea species parasitizing several small mammal species. My data show that *Bartonella grahamii*, *B. vinsonii*, *B. volans*, and an unidentified genetic lineage that may represent a new *Bartonella* species, are represented within these flea communities.

These results suggest that small mammals may be routinely exposed to *Bartonella*, which presumably increases risk of disease in mammalian communities that experience stress associated with ecological disruption.

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